

Almond Profit Function

Ixel Medrano

Sofia Sarak

Our almond profit function uses the parameters:

- price per pound that almonds are sold at
- cost per acre to grow almonds

The almond yield function we had created for Assignment 3 is embedded within our new profit function.

Set Up

```
# Load necessary libraries
library(tidyverse)
library(here)
library(janitor)

# Read in data
climate <- read_table(here("week 3/clim.txt")) %>%
  clean_names()
```

Test Case: Figuring Our Profit Function

```
# Source yield function
source(here("alm_yield.R"))

# Use yield function to find yield
yield <- alm_yield(climate)
```

```

[1] -0.3552237  9.2906757  68.9130633  15.4280698  20.2083803
[6]  2.4820009 1919.9811511   3.5818399 329.6938750  27.8636956
[11] -0.1436364  9.5999883 159.5119587   0.2450914 -0.2585997
[16] -0.2367722 656.3724121  18.6324135  20.2007396 576.2821943
[21]  0.7367438 153.7655092

```

```

# Assign values to our parameters of interest
price_lb <- 1.99
price_ton <- price_lb * 2000

cost_acre <- 3029

# Calculate revenue and profit
revenue <- price_ton * yield
profit <- revenue - cost_acre

```

Once we confirmed that our function and calculations are successful, we incorporated them into our larger function.

Building the Function

```

alm_profit <- function(price_lb, cost_acre, climate_data){

  ## Creating alm_yield function
  alm_yield <- function(climate_data){

    # Find mean minimum temperature in February of each year
    climate_month <- climate_data %>%
      filter(month == 2) %>%
      group_by(year) %>%
      summarize(min_temp_feb = mean(tmin_c))

    # Find total precipitation in January of each year
    precip_month <- climate_data %>%
      filter(month == 1) %>%
      group_by(year) %>%
      summarize(precip_sum_jan = sum(precip))

    # Combine to keep year -- important for almond profit calculation
    result <- climate_month %>%

```

```

    left_join(precip_month, by = "year")

    # Apply formula
    result$yield <- (-0.015 * result$min_temp_feb) - (0.0046 * (result$min_temp_feb)^2) -
      (0.07 * result$precip_sum_jan) + (0.0043 * (result$precip_sum_jan)^2) + 0

    return(result %>% select(year, yield))
  }

  ## Calculate profit
  result <- alm_yield(climate_data)

  # Calculate price per ton
  price_ton <- price_lb * 2000

  # Calculate revenue and profit
  result$profit <- (price_ton * result$yield) - cost_acre
  result$price_lb <- price_lb
  result$cost_acre <- cost_acre

  return(result)
}

```

The parameter values that we use in our test run as well as for benchmarks in our sensitivity analysis are 1.99 dollars per lb of almonds and 3029 dollars for cost to grow an acre. These values come from [Terrain's 2026 Almond Production report](#), and are averages.

We opted to keep our sensitivity analysis rather simple, and assume that these parameters stay constant over time.

```

# Apply function to test values
test <- alm_profit(price_lb = 1.99, cost_acre = 3029, climate)

```

This was successful!

Sensitivity Analysis

First, we create vectors of our possible parameter values. For our analysis, we choose 20 values that are equally-spaced, with our test values (defined above) as the center.

```
# Create vector of potential price per lb values
price_lb <- seq(1, 3, length.out = 20)

# And vector of cost per acre
cost_acre = seq(2000, 4000, length.out = 20)

# Combine into single parameters data frame
params <- cbind.data.frame(price_lb, cost_acre)
```

From there, we apply our `alm_profit` function and produce a data frame of `price_lb`, `cost_acre`, `year`, and `profit` values. This is our sensitivity analysis!

```
# Apply profit function to parameters data frame
results <- params %>%
  pmap(alm_profit, climate_data = climate)

# Create a data frame from results
results_df <- bind_rows(results, .id = "run_id")

# Add parameters to data frame
results_df <- results_df %>%
  left_join(params %>% mutate(run_id = as.character(row_number())) , by = "run_id")
```

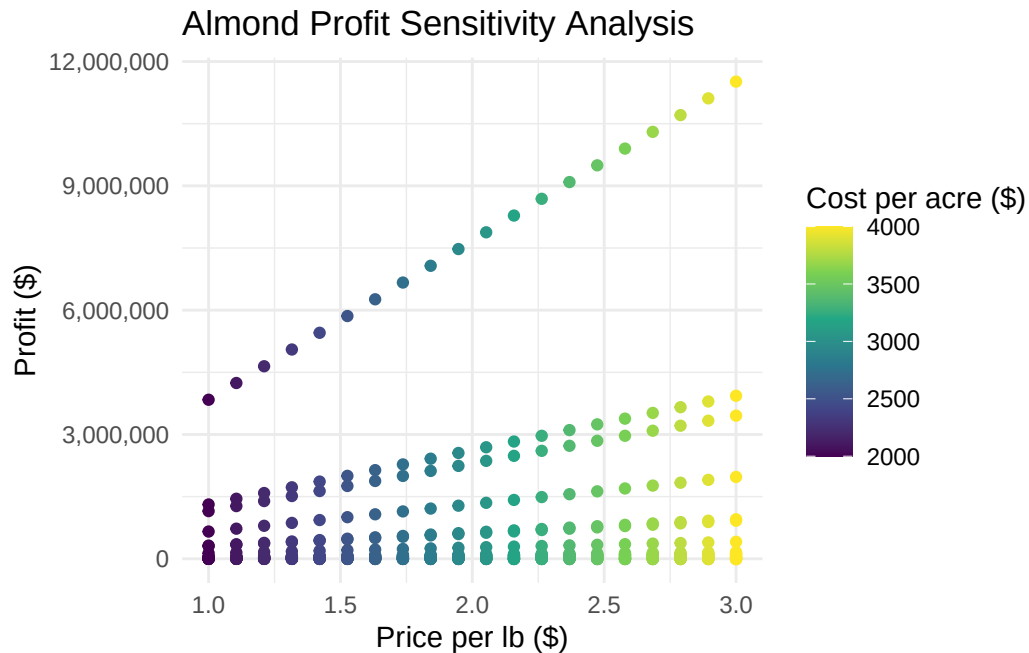
Visualizing Results

Plotting the sensitivity of almond profit to almond price per lb as well as cost per acre.

```
# Initialize ggplot, assign each variable of interest to different aspect of graph
ggplot(results_df, aes(x = price_lb.x, y= profit, color = cost_acre.x)) +

  # Scatter plot
  geom_point() +

  # Labels and customizations
  labs(title = "Almond Profit Sensitivity Analysis",
       x = "Price per lb ($)",
       y = "Profit ($)",
       color = "Cost per acre ($")) +
  theme_minimal() +
  scale_color_viridis_c() +
  scale_y_continuous(labels = scales::comma)
```



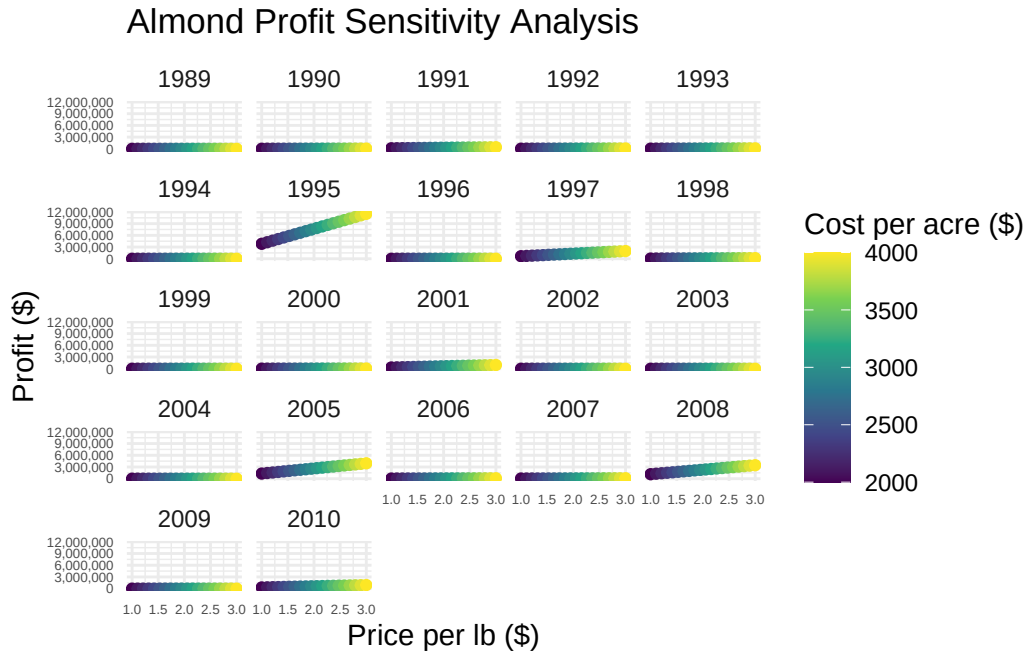
Doing the same, but faceting by year to see how changes in yield over time affect sensitivity.

```
# Initialize ggplot, assign each variable of interest to different aspect of graph
ggplot(results_df, aes(x = price_lb.x, y = profit, color = cost_acre.x)) +

# Scatter plot
geom_point() +

# Labels and customizations
labs(title = "Almond Profit Sensitivity Analysis",
      x = "Price per lb ($)",
      y = "Profit ($)",
      color = "Cost per acre ($)") +
theme_minimal() +
theme(axis.text = element_text(size = 5)) +
scale_color_viridis_c() +
scale_y_continuous(labels = scales::comma) +

# Facet by year
facet_wrap(~year)
```



Conclusion

Based on our analysis, it appears that almond price per lb has a larger effect on almond profit compared to cost per acre. Higher costs per acre can still be associated with relatively high profits. Perhaps this is because the scale of our variables is so different: a 10-cent difference in price per lb may be a lot more impactful than a 100-dollar change in cost per acre.

Our faceted plot shows us that the relationship between our three variables does change depending on yield (though, it is a bit difficult to make out distinctions; the large number of years squishes all of our data). Also, it can be noted that we see a linear pattern because of the way we chose to simulate our `pice_lb` and `cost_acre` variables. Using a distribution function like `rnorm()` could potentially yield very different results.